Introduction to ADaM data structures

- In clinical trials, data specified in the protocol is collected using case report forms
- The data collected using case report forms is called raw data
- Raw data is then transformed into CDISC compliant SDTM datasets
- SDTM data is used as input to create analysis datasets
- Analysis datasets compliant with CDISC ADaM standard are called ADaM datasets
- CDISC ADaM standard has three standard dataset structures
    - ADSL- Subject Level Analysis Dataset
    - BDS - Basic Data Structure
    - OCCDS - Occurrence Data Structure

Introduction to BDS structure

- The typical record structure in a BDS dataset is 'one or more records per subject, per analysis parameter, per analysis timepoint'
- An easy phrase to remember the structure of BDS is 'a dataset that is used to store the results of tests'
- A 'Result' is formally called an 'analysis value'
- A 'test' is formally called as 'analysis parameter'
- A 'test' could be conducted at different timepoints. Hence, we have the concept of 'analysis timepoint' as well.
- So, there are core variables prescribed in ADaM standard for 'analysis parameter' and 'analysis value'
- Variables for 'analysis parameter' are PARAMCD, PARAM, PARAMN
- Important variables for 'analysis result' are AVAL, AVALC.
- Variables for baseline value (BASE), change from baseline (CHG), and percent change from baseline (PCHG) are also considered as analysis values
- Similar to variables for 'analysis parameter' and 'analysis value', there are standard variables prescribed for 'analysis timepoint'
- Variables for 'analysis timepoint' include AVISIT, AVISITN, ADT, ATM, ADTM, ADY
- Here is a visualization of some variables for data related to vital signs
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USUBJID	PARAMN	PARAMCD	PARAM	AVISIT	ADT	AVAL
CSG-1001	1	STSBP	Sitting Systolic Blood Pressure (mmHg)	Week 1	15APR2003	122
CSG-1001	2	STDBP	Sitting Diastolic Blood Pressure (mmHg)	Week 1	15APR2003	74

Introduction to rules governing addition of new parameters

- When creating analysis datasets, we typically run into the question of how to structure the derived information - whether to add rows or columns.
- ADaM implementation guide provides certain rules that guide on when to add the information as rows and when to add the information as columns
- Within the concept of rows - we end up with two options
    - Add the derived information as rows within existing parameter
    - Add the derived information as rows as a new parameter
- Of the 6 rules available, 3 rules are related to addition of new parameters
    - 2. **Add additional derived data as needed for the analysis.** The second step consists of adding derived rows and columns based on the initial set of ADaM dataset records and columns. ***The rules below govern this step***. These rules are further described and illustrated in the remaining subsections of this section.

      Rule 1: A parameter-invariant function of AVAL and BASE on the same row that does not involve a transform of BASE should be added as a new column.

      Rule 2: A transformation of AVAL that does not meet the conditions of Rule 1 should be added as a new parameter, and AVAL should contain the transformed value.

      Rule 3: A function of one or more rows within the same parameter for the purpose of creating an analysis timepoint should be added as a new row for the same parameter.

      Rule 4: A function of multiple rows within a parameter should be added as a new parameter.

      Rule 5: A function of more than one parameter should be added as a new parameter.

      Rule 6: When there is more than one definition of baseline, each additional definition of baseline requires the creation of its own set of rows.